

03

EASYRIDE: TECH AT DRIVING

Product Design

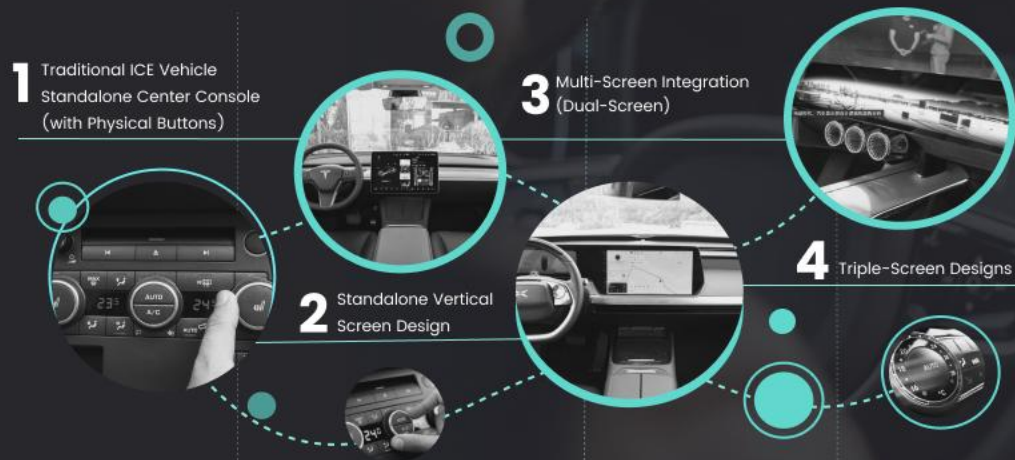
Innovative Design

As Vehicles Become Increasingly Digital, The Loss Of Physical Buttons Alienates Users Who Value Tactile Feedback And Intuitive Control. This Project Reintroduces Modular Physical Buttons For Electric Vehicles, Creating A Safer And More Inclusive Driving Experience For Older Adults And Those Resistant To Touchscreen Interfaces.

Year : 2024
Individual Project

BACKGROUND

EV Dashboard Design: From Buttons To Multi-Screens



Field Research

I visited Tesla and Lixiang offline store for a field research to better understand the center console design of electric vehicles.



Unresponsive voice recognition system

• Problem

Voice assistant often fails to recognize user commands accurately, especially in noisy driving environments



Overly complex central control interface

• Problem

Climate controls are buried in multiple submenus
Lack of clear hierarchy or prioritization for frequently used functions (e.g., AC, navigation).



Lack of haptic feedback

• Problem

Users cannot confirm interactions by touch alone, increasing visual dependency.

The Issues With Touchscreen In EVs

Climate controls are buried in multiple submenus

Overly Complex Menu Hierarchy

Opening "AC main interface" > navigating tabs

Increased Driver Distraction

Screen operations take longer than physical buttons

Reliance on touchscreen with no physical buttons/knobs

Impossible to Operate Blindly

Drivers must visually focus on the screen

Inefficient Emergency Adjustments

Critical functions (e.g., defogging, max cooling) require multi-level menu access.

Touchscreen Distraction: Riskier Than Drunk Driving?

The chart shows the percentage increase in distracted drivers' response times. An undistracted driver typically reacts in 1 second.

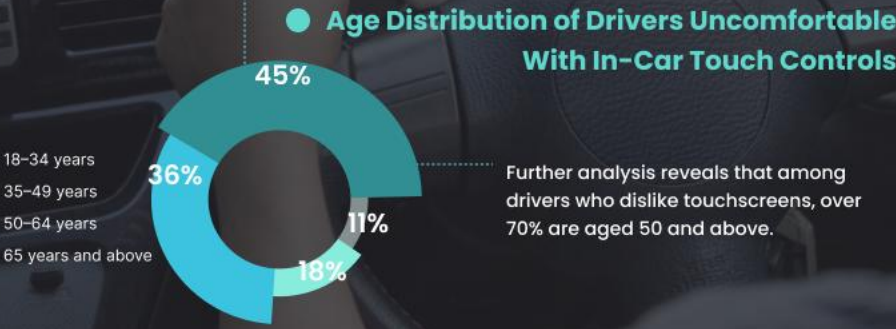


Excessive use of touch screens can dangerously distract drivers, studies suggest it's more hazardous than drunk driving.

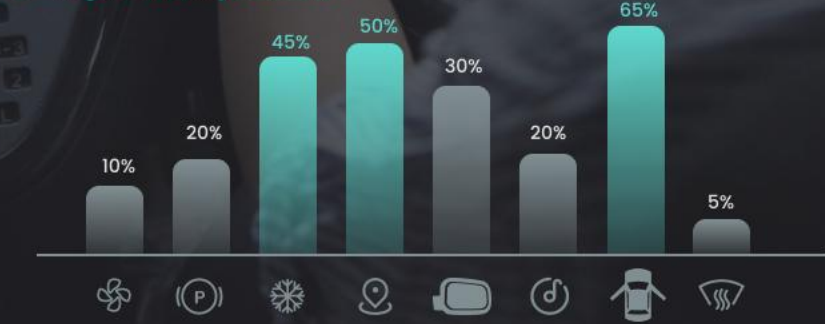
USER RESEARCH

Drivers' Attitude Toward In-Car Touchscreens

This research aims to explore drivers' attitudes toward in-car touchscreen systems and identify the user groups that struggle most with adapting to digital interfaces, providing a foundation for designing alternative physical control solutions.



High-Priority Buttons



Interview

Mr. Wang

Age: 53

Male

Retiree

Painpoint

That central control screen is filled with tiny icons and I can never figure out how to use it properly.

Need

The biggest problem for me is the lack of physical controls.

Wants

Hmm... another thing is to keep it simple, not too complicated.

Need

It'd be great if there were a "simple mode" that you could use as soon as you get in the car, without having to fuss around.

Mrs. Li

Age: 62

Female

Retiree

Shawn

Age: 32

male

Office Worker

Wants

Also, could the text on the screen be bigger? And the icons too, with stronger color contrasts, so it's easier for us elderly to see clearly.

Need

Also, I'd like a big, easy-to-reach button for emergencies, like calling for help if I'm feeling unwell.

Conclusion

The degree of inability to operate the central control screen proficiently is positively correlated with the age of the car owner.

User Journey Map

Control The Air Conditioner

- Painpoints :
- Requires a great deal of attention.
 - Requiring repeated screen taps to adjust temperature.
- Opportunities :
- Clear and visible Buttons

Check The Central Control Screen

- Painpoints :
- Icons and text on the screen are too small and hard to read.
 - Menu hierarchy is complex.
- Opportunities :
- Larger Icons and Fonts

Turn On The Defogging Button

- Painpoints :
- Don't know where to click the defogging button.
 - Don't know which one is the defogging button.
- Opportunities :
- Icons and text prompt together

Arrive At The Destination

- Painpoints :
- Not familiar with the automatic parking function.
 - Forgot to shift gears and pull the handbrake.
- Opportunities :
- There are corresponding safety tips

Set Up Navigation

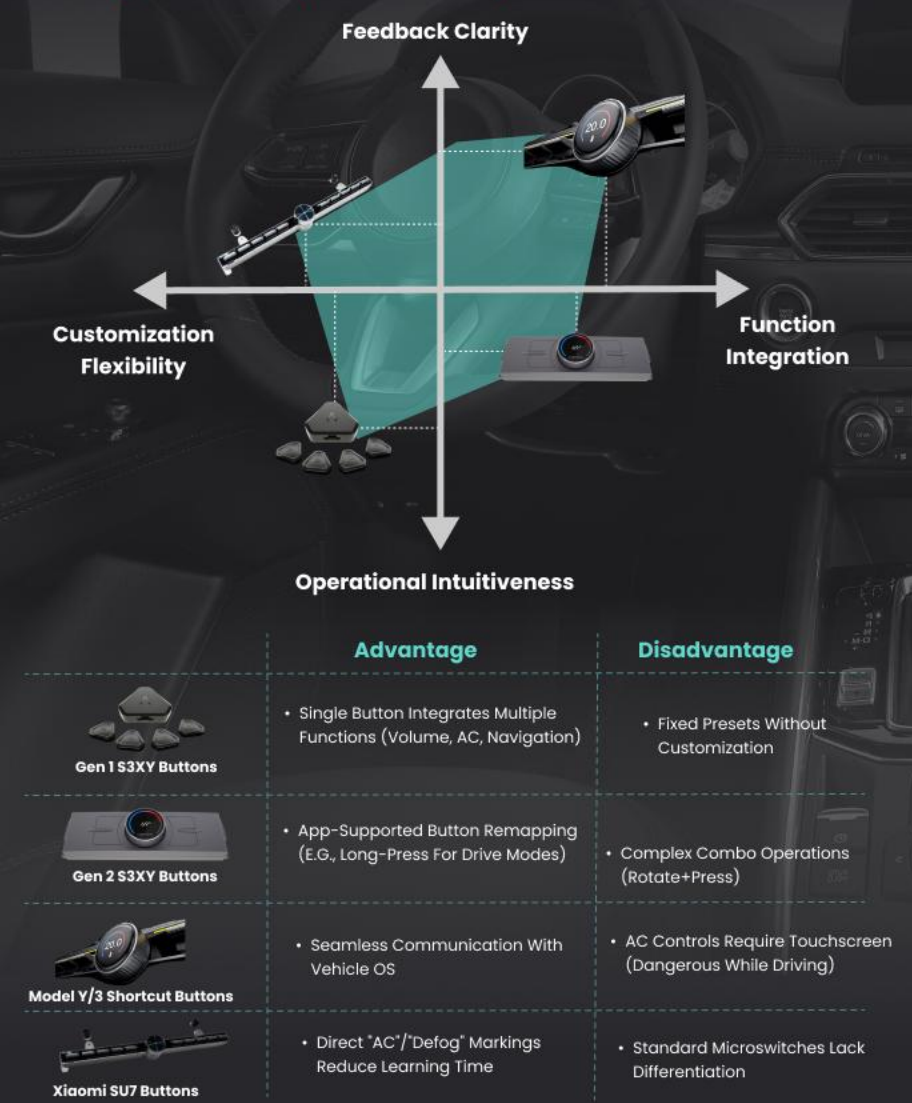
- Painpoints :
- Complex navigation interface with unclear route instructions.
- Opportunities :
- Simpler navigation function.

Findings

- Due to the decline in the tactile perception of elderly users, their touch perception of the screen is not very precise
- Due to the decline in memory ability of the elderly, they often forget the operation logic of the screen and some safety knowledge

COMPETITOR STUDY

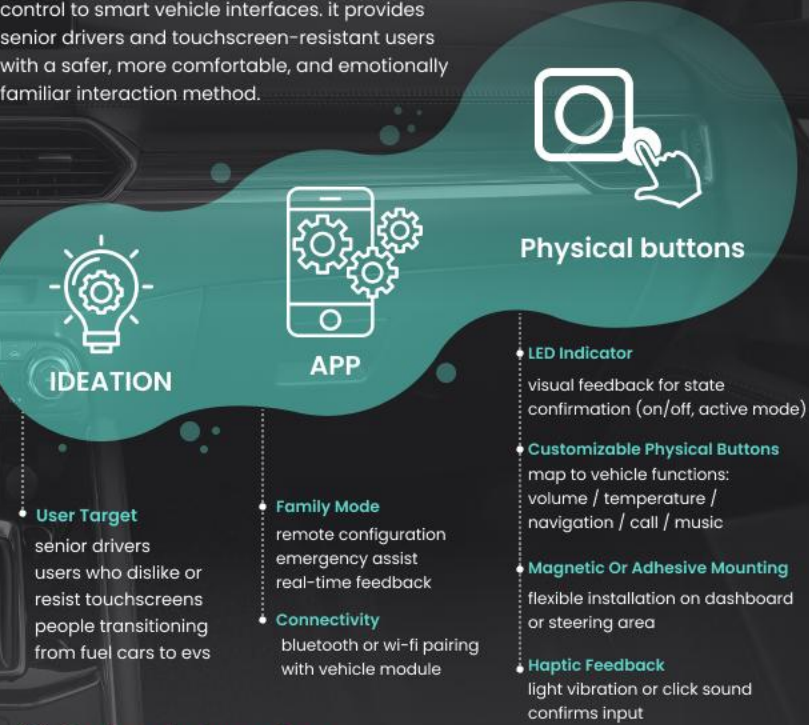
Competitive Analysis Matrix: Physical Control Interfaces In EVs



IDEATION

Mind Map

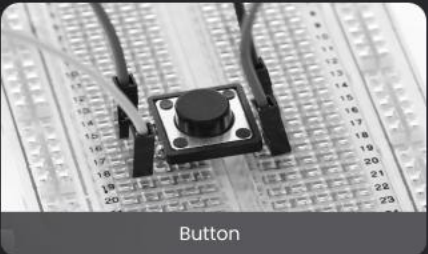
The project aims to bring back tangible, intuitive control to smart vehicle interfaces. it provides senior drivers and touchscreen-resistant users with a safer, more comfortable, and emotionally familiar interaction method.



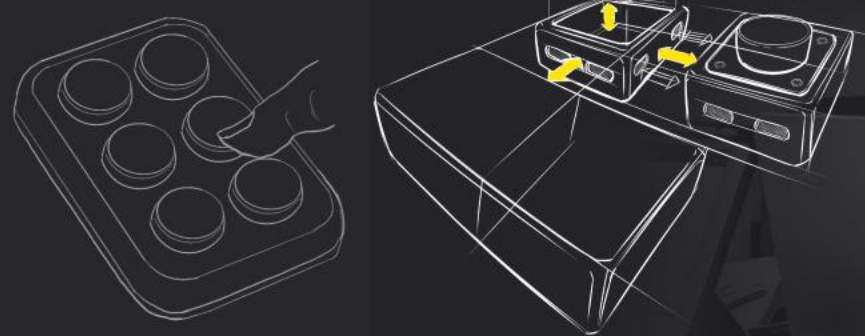
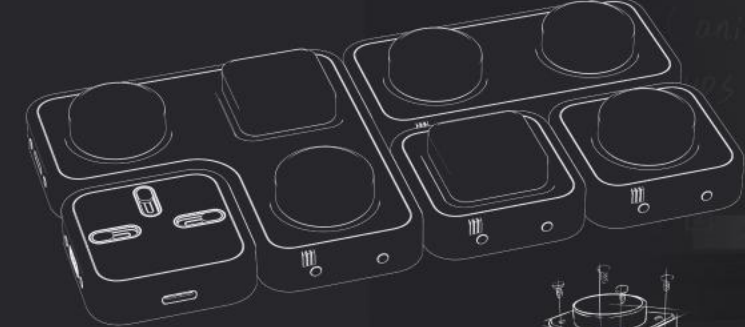
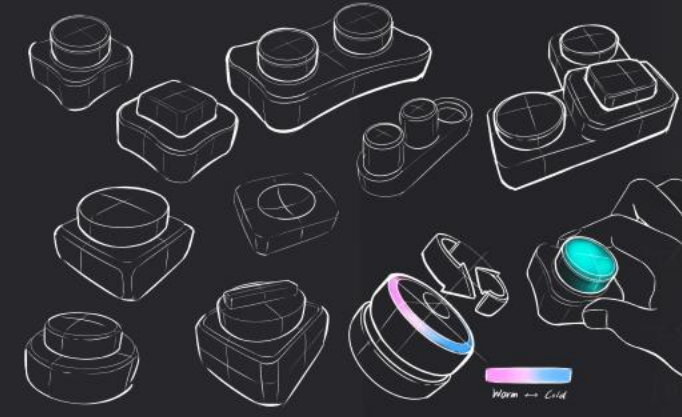
System Integration



Intention Map



Sketching



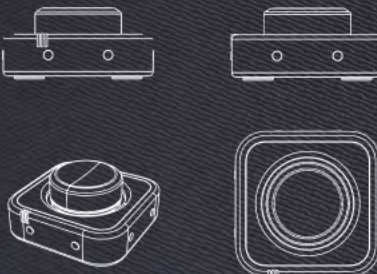
BRAINSTORMING



DELIVER

Function

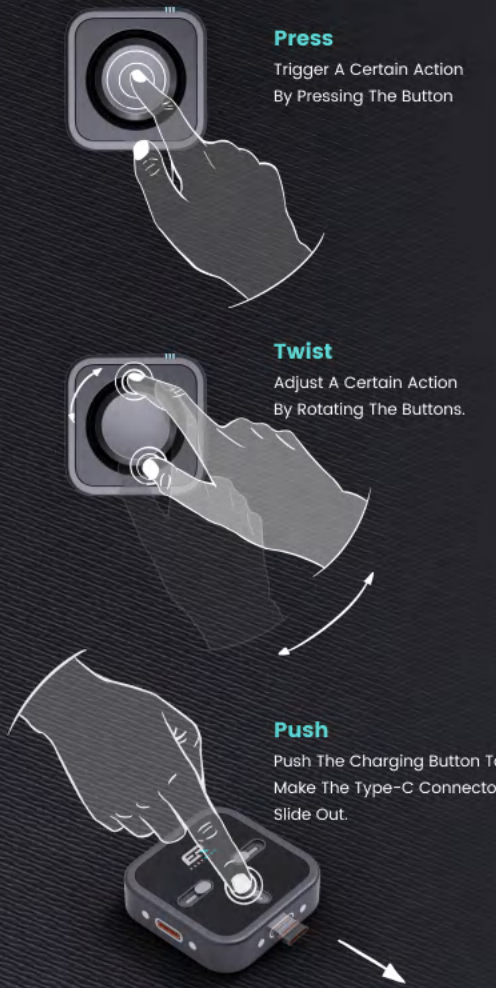
Single Button Integrates Multiple Functions (Volume, AC, Navigation)Single Button Integrates Multiple Functions (Volume, AC, Navigation)Single Button Integrates Multiple Functions (Volume, AC, Navigation)Single Button Integrates Multiple Functions (Volume, AC, Navigation)



Specification



Interactive Gestures



DELIVER

Details



USAGE & SCENARIOS

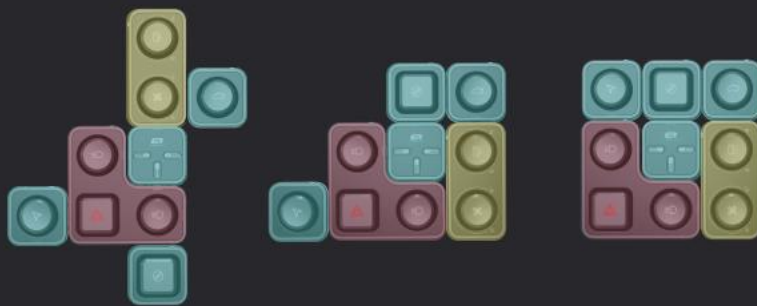
Scenarios For The Use Of Freedom

The back of the product is coated with adhesive, allowing it to be easily attached to any position within the vehicle for convenient and easy access.



LET YOUR BUTTONS BE ARRANGED FREELY !

The size of the product is standardized. There are magnets on all four sides of the product. Regardless of the size, the keys of any type can be magnetically attracted to each other and closely combined, making the product more personalized.



Method One

Method Two

Method Three

Branding

Logo Design

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Colour



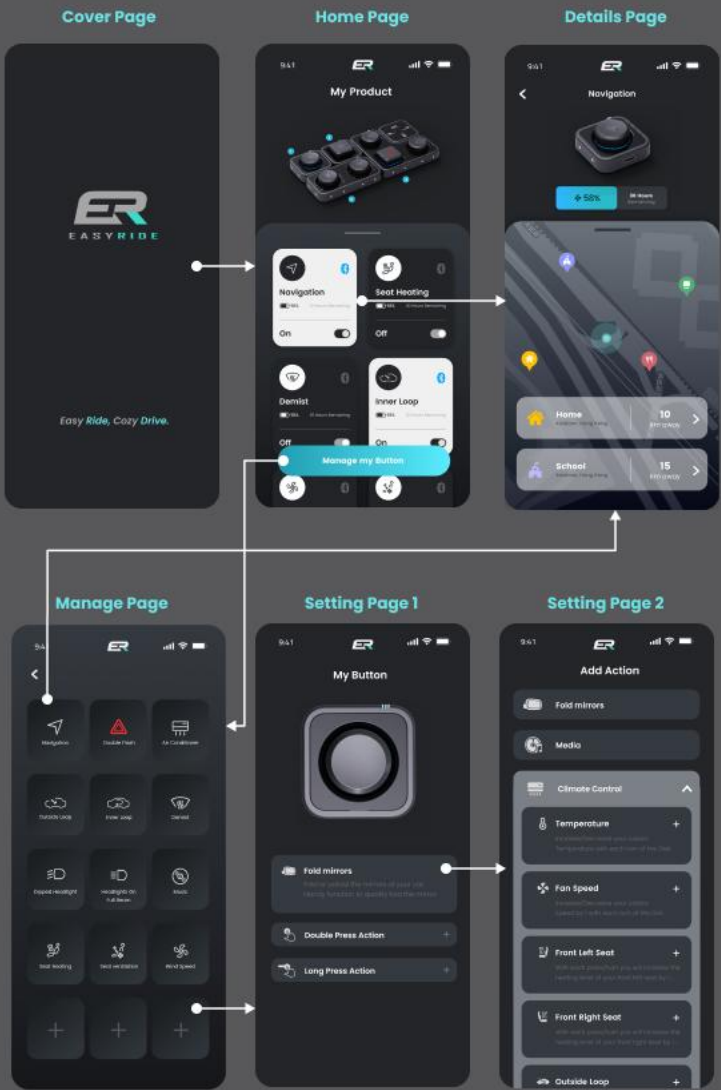
Font

Poppins Bold Poppins SemiBold Poppins Medium Poppins Regular

Icon



Mobile App "Easyride"



FINAL OUTCOME

